How fast is fast enough? The onset-to-isolation delay in the 2026 Bundibugyo outbreak, DRC

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Bottom line. If isolation of symptomatic cases is 80% effective, cutting the mean delay from onset to isolation from 7 to 5 days would avert about 73% of the next 12 months' cases and from 7 to 3 days about 89%, turning a plateau ($R_e \approx 1.00$) into decline. Even at 50% effectiveness a 3-day delay averts 75%. The effect is large because the outbreak sits at the epidemic threshold, and it reverses if the true current delay is already under 5 days, so the delay must be measured.

Question and approach

By 12 September 2026 the DRC had 7,200 confirmed cases and a modelled incidence of about 83 new cases per day. We asked how many cases would be prevented, and how much sooner transmission would end, if the mean delay from symptom onset to isolation fell from 7 to 5 or 3 days, with isolation that is 50%, 80% or 100% effective (reducing transmission of isolated cases by that fraction). We used a deterministic susceptible–exposed–infectious–isolated–removed (SEIQR) model (Fig. 1) with isolation rate $\tau = 1/\text{delay}$; isolated people transmit at $1-\epsilon$ of the rate of others, so $R_e = (\beta/\gamma)\{1 - \epsilon\tau/(\tau+\gamma)\}$ (details in the manuscript appendix). The model was fitted by maximum likelihood to confirmed cases by report date up to 12 September [1], allowing one change in transmission (cross-checked against the onset curve corrected for reporting delay). It fitted well (95% predictive coverage 95%). Transmission grew (R_e 1.20) until about 24 July and has since plateaued (R_e 1.00 (0.93 to 1.06)).

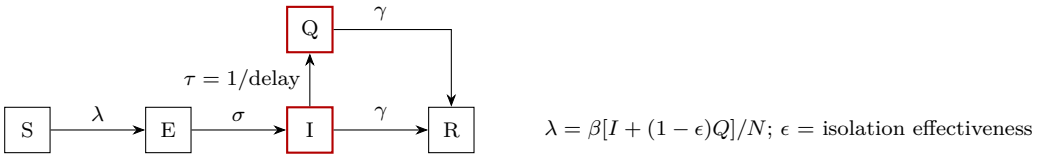


Figure 1: Model structure; the intervention is the flow $I \to Q$.

Scenarios and uncertainty

Nine scenarios (delay 7, 5, 3 days × ϵ 50, 80, 100%) start on 12 September 2026. Each of 1,000 paired draws combines the fitted growth rate and incidence with latent (mean 6.3 d, range 4.0–10.1) and infectious periods (mean 7 d, range 5–10) drawn from triangular distributions [2, 3]; transmission is calibrated so the 7-day baseline reproduces the fitted growth, and only the delay changes between scenarios. Outcomes: cumulative symptomatic cases over 12 months, cases averted versus the 7-day delay, and months until fewer than one exposed or infectious person remains. We also ran a stochastic version (500 replicates), partial-rank-correlation and 17 structural sensitivity analyses, and an out-of-sample forecast check. All code is in R; the repository regenerates every number and figure.

Results

Table 1: Results (median and 95% uncertainty interval, 1,000 paired draws; † cases averted × 48%, illustrative).

Isolation effectiveness	Delay (days)	R_e median (95% UI)	Cases in 12 months median (95% UI)	Cases averted (%) median (95% UI)	Deaths averted† median	Months to end median (95% UI)	P(ended ≤24 mo)
50%	7	1.00 (0.92–1.08)	31,201 (9,854–109,025)	–	–	>60	0.01
	5	0.95 (0.86–1.02)	15,405 (6,232–44,057)	50 (33–62)	7,532	53 (21 to >60)	0.05
	3	0.87 (0.80–0.93)	7,695 (4,033–16,443)	75 (57–86)	11,346	22 (13 to 49)	0.60
80%	7	1.00 (0.92–1.07)	31,089 (9,845–107,831)	–	–	>60	0.01
	5	0.89 (0.82–0.95)	8,212 (4,127–18,244)	73 (55–84)	11,023	24 (14 to 53)	0.52
	3	0.74 (0.68–0.79)	3,313 (2,133–5,309)	89 (76–95)	13,421	9 (7 to 14)	1.00
100%	7	1.00 (0.93–1.06)	30,961 (9,833–106,530)	–	–	>60	0.01
	5	0.83 (0.77–0.88)	4,807 (2,779–8,746)	84 (68–93)	12,588	13 (9 to 21)	0.99
	3	0.60 (0.55–0.64)	1,817 (1,234–2,728)	94 (85–98)	14,050	5 (4 to 7)	1.00

With a 7-day delay the model projects 31,089 (9,845 to 107,831) cases over 12 months and transmission that persists beyond five years. At 80% effectiveness a 5-day delay lowers R_e to 0.89 and a 3-day delay to 0.74, with transmission ending after a median of 24 and 9 months. At 50% effectiveness the corresponding times are 53 and 22 months; at 100% they are 13 and 5 months. Returns diminish: the first two days saved (7 to 5) avert more than the next two.

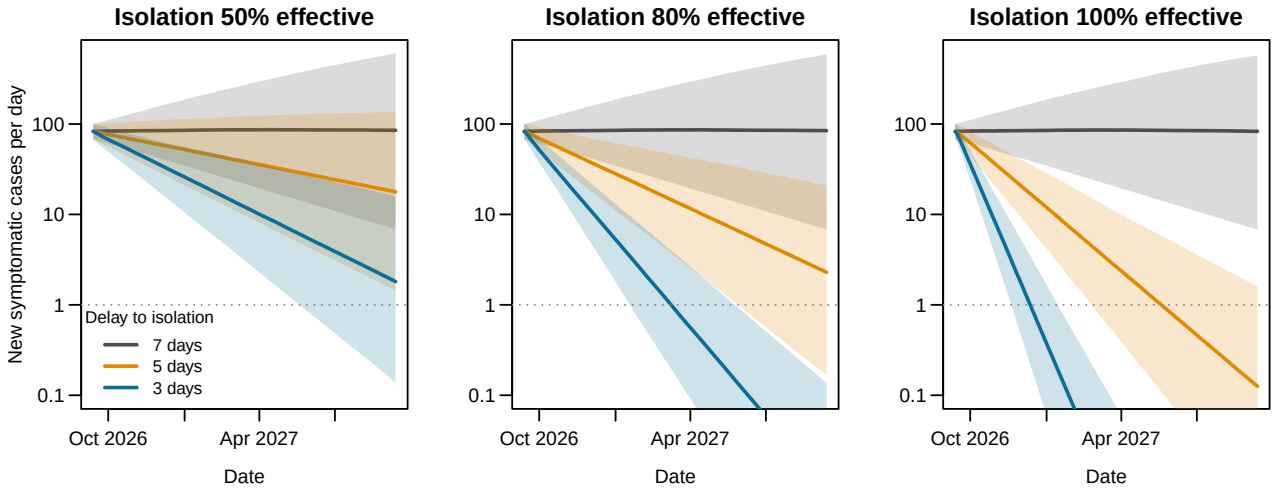


Figure 2: Projected new symptomatic cases per day (median and 95% UI, log scale) from 12 September 2026.

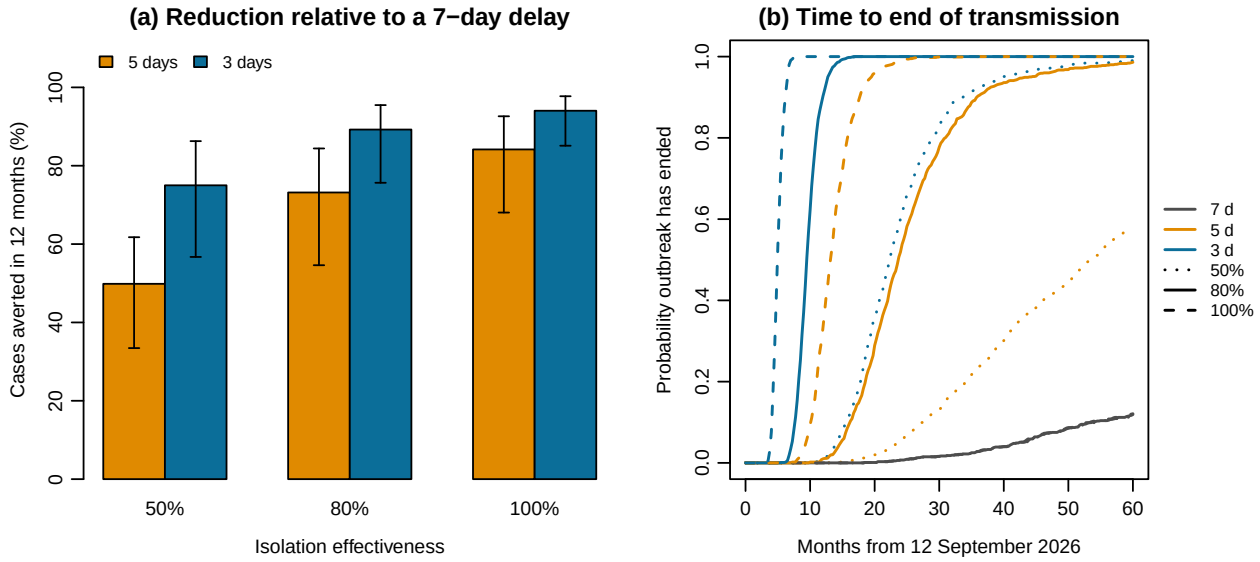


Figure 3: (a) Percent of 12-month cases averted relative to a 7-day delay. (b) Probability transmission has ended, by months.

How robust is this?

The stochastic model gave the same medians (ratio 1.00–1.09). The isolation delay (partial rank correlation 0.94) and effectiveness (−0.80) dominate the uncertainty, followed by baseline growth (0.42). Across assumptions (natural history, Erlang-distributed delays, delay defined among isolated cases, 90-day roll-out) the 5-day benefit stayed between 56 and 84% and the 3-day benefit between 77 and 96%. **The exception is the baseline delay:** if the true current delay were 4.6 days, a "5-day" target would increase cases (percentage averted: −67 (−98 to −37)). A forecast using data to 12 September overestimated the next 46 days of cases by 23%, so absolute counts should be read as conditional on the plateau.

Decision relevance and limits

Shortening the delay (community alert networks, decentralised sampling, rapid results, nearby isolation capacity) is worth the most when isolation is also effective; a quick delay with poor isolation helps less. Percent reductions apply while the outbreak is near $R_e \approx 1$; at higher R_e a 3-day delay alone suffices only if baseline $R_e \leq 1.36$ (80% effectiveness). Limits: the 7-day baseline comes from the project brief and is not confirmed; isolation effectiveness is assumed, not estimated; infectiousness is constant and contact tracing, funerals and spatial structure are omitted; national data are treated as homogeneous. Code and data: repository README (URL to be added).

References

[1] Sam Abbott, Katharine Sherratt, et al. BVDOutbreakSize: analysis and cleaned surveillance streams for the 2026 Bundibugyo virus disease outbreak. https://github.com/epiforecasts/BVDOutbreakSize, 2026. Data as of 12 September 2026; underlying sources: INSP situation reports and INRB-UMIE transcriptions.

[2] Dav M. Ebengo, Pierre Akilimali, et al. Early epidemiological and spatial assessment of the Bundibugyo Ebola virus outbreak 2026. Epidemiological forum (preprint, not peer reviewed), 2026. Posted 12 June 2026.

[3] Gerardo Chowell and Hiroshi Nishiura. Transmission dynamics and control of Ebola virus disease (EVD): a review. *BMC Medicine*, 12:196, 2014. doi: 10.1186/s12916-014-0196-0.